Criterion 2: Outcome Based Teaching Learning (120)

by
Dr. T V Rama Krishna
Professor of ECE

2.1. Describe Processes Followed to Ensure Quality of Teaching & Learning (20)

- **Adherence to Academic Calendar**
  - **Academic activities are planned in line with the University/Institute calendar ensuring timely commencement and completion of courses.**
  - **Departmental academic calendars are prepared with detailed schedules for theory, labs, assessments, seminars, and remedial sessions.**
  - **Lesson plans and course files prepared by faculty are mapped with calendar weeks and reviewed periodically.**
  - **Monitoring mechanism: HoD/IQAC/Academic Council conducts checks to ensure teaching–learning is on schedule.**
- **Instructional & Pedagogical Initiatives**
  - **Faculty adopt student-centered pedagogies such as problem-based learning, case studies, and flipped classrooms.**
  - **Use of real-world examples and industry-oriented case studies to link theory with practice.**
  - **Collaborative learning promoted through group discussions, peer-to-peer learning, and team projects.**
  - **Integration of ICT tools (LMS, smart classrooms, simulation software, online labs, MOOCs).**

- **Laboratory Teaching–Learning Quality**
  - **Well-structured lab manuals containing objectives, procedures, observation formats, and analysis methods.**
  - **Students are trained to conduct experiments systematically, record observations accurately, analyze data, and draw conclusions.**
  - **Continuous improvement through lab audits, student feedback, and faculty review meetings.**
  - **Use of virtual labs, industry software, and simulation platforms to enhance experimental learning experience.**
- **Catering to Learner Diversity**
  - **Slow learners: remedial classes, bridge courses, doubt-clearing sessions, mentoring, peer-learning support.**
  - **Fast learners: advanced assignments, open-ended problem-solving, research internships, industry projects, NPTEL/Coursera enrollments.**
  - **Encouragement of innovation and entrepreneurship activities (hackathons, start-up cells, mini-projects).**
  - **Mentorship system ensures continuous tracking of both academic and personal development.**

- **Assessment of Learning & Attainment**
  - **Continuous Internal Evaluation (CIE): mid-semester tests, assignments, quizzes, viva voce, seminar evaluation.**
  - **End-Semester Evaluation (ESE): university-level written and practical exams.**
  - **Rubrics-based assessments for labs, projects, and presentations to ensure transparency and objectivity.**
  - **Periodic CO → PO/PSO attainment analysis to measure effectiveness of teaching–learning.**
- **Implementation Details & Documentation**
  - **Course files, lab manuals, academic calendars, and lesson plans serve as primary evidence.**
  - **Minutes of departmental meetings, feedback reports, IQAC reports, and academic audits maintain institutional record.**
  - **Remedial & advanced learner logs, mentorship records, and learning outcome attainment reports demonstrate process adherence.**
- **Monitoring & Feedback Mechanism**
  - **Student feedback collected (mid/end-semester) on course delivery, clarity, pace, and engagement.**
  - **Class committee meetings and open forums allow students to share difficulties and suggestions.**
  - **Corrective actions (pedagogical changes, mentoring sessions, additional tutorials) are documented with Action Taken Reports (ATR).**
  - **Peer review and academic audits provide external quality checks.**

- **Impact Analysis**
  - **Increased student pass percentage and progression rates.**
  - **Enhanced practical and analytical skills through quality laboratory training.**
  - **Stronger industry readiness and employability due to real-world examples and project-based learning.**
  - **Evidence of improvement in CO/PO attainment, placement statistics, research output, and student satisfaction levels.**

Marks Distribution (20):

- **Adherence to Academic Calendar & Lesson Planning → 5 Marks**
- **Pedagogical & Student-Centric Initiatives (real-world, collaborative, ICT, lab quality) → 5 Marks**
- **Student Diversity Support (slow & fast learners, mentoring) → 5 Marks**
- **Documentation & Impact Analysis → 5 Marks**

2.2. Quality of Student Capstone Project (25)

1. Project Identification and Allotment

- **Process of Selection**
  - **Students encouraged to identify projects based on real-world problems, industry needs, and research gaps.**
  - **Ideas sourced from industry collaboration, research centers, hackathons, societal challenges, and innovation cells.**
  - **Projects categorized into application-oriented, product development, research, or review-based.**
- **Project Allotment**
  - **Faculty guides allocated based on specialization, research area, and industry experience.**
  - **Interdisciplinary projects encouraged, aligning with NEP-2020's emphasis on cross-disciplinary learning.**
  - **Project proposals vetted by Departmental Project Review Committee (DPRC) for feasibility and relevance.**

2. Consideration of Quality Parameters

- **Environment & Sustainability → Projects addressing eco-friendly designs, renewable energy, waste management, or sustainable products.**
- **Safety → Compliance with safety standards in labs/industry-based projects.**
- **Ethics → Ethical handling of data, plagiarism-free reports, responsible innovation.**
- **Cost → Focus on cost-effectiveness and affordability of proposed solutions.**
- **Standards → Alignment with national/international standards (ISO, IEEE, BIS).**
- **Type of Project → Balanced distribution among application, product, research, and review projects.**

3. Monitoring and Evaluation Process

- **Continuous Monitoring**
  - **Regular progress reviews conducted by Faculty Guide and DPRC.**
  - **Mid-term presentations, status reports, and progress seminars organized.**
  - **Evaluation rubrics framed covering innovation, technical content, feasibility, and relevance.**
- **Prototype Development & Demonstration**
  - **Students must present working prototypes/models/software tools where applicable.**
  - **Projects validated through demonstrations, testing, and peer/industry expert evaluation.**
- **Final Evaluation**
  - **Conducted by internal faculty + external examiner (industry/research expert).**
  - **Assessment covers: report quality, viva voce, prototype demonstration, and societal relevance.**

4. Relevance and Impact of Projects

- **Projects linked to industry problems, community challenges, or cutting-edge research.**
- **Students exposed to entrepreneurship opportunities via innovation cells and incubation centers.**
- **Evidence of outcomes:**
  - **Publications, patents, product launches, or start-ups.**
  - **Awards/recognition in competitions or conferences.**

5. Implementation Details: PO/PSO Mapping

- Each project mapped to Program Outcomes (POs) and Program Specific Outcomes (PSOs).
- Example Mapping:
  - PO2 (Problem Analysis): Students analyze complex engineering problems.
  - PO3 (Design/Development): Projects develop solutions considering safety, sustainability, and standards.
  - PO7 (Environment & Sustainability): Projects reflect environmental responsibility.
  - PO8 (Ethics): Students practice professional ethics in data handling, plagiarism checks.
  - PSOs: Address discipline-specific problem solving (e.g., AI/ML algorithms, IoT applications, Power Systems design).
- Justification: Mapping documented in project records, with evidence of how each PO/PSO is attained.

6. Evidence and Documentation

- Project Proposals, DPRC Review Minutes, and Allotment Records.
- Progress Seminar Reports and Monitoring Sheets.
- Prototype Demonstration Videos/Photos.
- Final Evaluation Reports, External Examiner Assessments.
- CO–PO–PSO Attainment Mapping Sheets (Project Level).
- Student Achievements (patents, publications, awards).

Marks Allocation (25):

- Project Identification & Relevance → 5 Marks
- Quality Factors (Environment, Sustainability, Safety, Ethics, Cost, Standards) → 5 Marks
- Monitoring & Evaluation Process → 5 Marks
- Working Prototype/Research Relevance → 5 Marks
- PO/PSO Mapping & Justification → 5 Marks

2.3. Internship/Industrial Training (10)

1. Process Followed for Internship/Industrial Training

- **Identification of Opportunities**
  - **Students are encouraged to apply for internships through:**
    - **Institute's Training & Placement (T&P) Cell.**
    - **Industry tie-ups, MoUs, and collaborations.**
    - **Online internship portals and alumni networks.**
  - **Preference given to core companies, research labs, startups, and government organizations.**
- **Allotment & Approval**
  - **Students select an area aligned with their discipline and career goals.**
  - **Proposals are reviewed and approved by HoD / Internship Coordinator.**
  - **Internship Guide (faculty mentor) assigned for monitoring.**
- **Monitoring & Evaluation**
  - **Students maintain an internship diary/logbook recording tasks, skills learned, and outcomes.**
  - **Mid-term reviews conducted by faculty guide (via presentations/reports).**
  - **On completion, students submit:**
    - **Internship report.**
    - **Certificate from host organization.**
    - **Presentation/viva before departmental panel.**
- **Integration with Curriculum**
  - **Internship mapped as a credit course (typically 2–4 weeks, 2–4 credits depending on program).**

Sometimes integrated with mini-projects or capstone projects

2. Typical Duration

- **Second-Year Students (Pre-final exposure): 2–3 weeks (summer training/industrial visits).**
- **Third-Year Students (Skill-building internships): 4–6 weeks (core discipline focus).**
- **Final-Year Students (Capstone-linked internships): 8 weeks or full-semester internships.**
- **Duration varies as per AICTE model curriculum and university guidelines.**

3. POs and PSOs Addressed through Internships

- **PO1 (Engineering Knowledge): Application of classroom concepts in real-world scenarios.**
- **PO2 (Problem Analysis): Identification and analysis of industry-relevant problems.**
- **PO3 (Design/Development): Exposure to design constraints (cost, sustainability, safety).**
- **PO4 (Investigation of Complex Problems): Data collection, experiments, performance testing.**
- **PO5 (Modern Tool Usage): Use of industry-grade software, simulators, and tools.**
- **PO6 (Engineer & Society): Understanding societal, legal, and ethical aspects of industry practices.**
- **PO7 (Environment & Sustainability): Awareness of green technologies and sustainable operations.**
- **PO8 (Ethics): Professional ethics in workplace.**
- **PO9 (Individual & Team Work): Functioning as part of cross-functional teams.**
- **PO10 (Communication): Preparation of reports, presentations, and technical discussions.**
- **PO12 (Lifelong Learning): Adapting to new skills, technologies, and industrial practices.**
- **PSOs (Program Specific Outcomes):**
    - **For CSE: Hands-on with AI/ML tools, databases, cloud environments.**
    - **For ECE: Exposure to embedded systems, VLSI design, telecom industry practices.**
    - **For ME: Training in CAD/CAM, manufacturing processes, thermal/automotive industries.**
    - **For EE: Training in power plants, electrical distribution, renewable energy systems.**

4. Documentation & Evidence

- Internship circulars, allotment letters, and MoUs with industries.
- Internship logbooks, weekly reports, and certificates from host organizations.
- Student presentations and viva reports.
- Consolidated record of PO/PSO mapping with internships.

Marks Allocation (10):

- Process & Documentation → 3 Marks
- Duration & Curriculum Integration → 2 Marks
- POs/PSOs Mapping with Justification → 5 Marks

2.4. Seminar and Mini/Micro Projects (10)

1. Process of Conducting Seminars

- **Selection of Topics**
  - **Students choose seminar topics from emerging technologies, research areas, industry practices, or socially relevant issues.**
  - **Topics vetted and approved by faculty guide / departmental committee to ensure relevance.**
  - **Encourages self-learning, literature review, and critical analysis.**
- **Execution**
  - **Each student/team prepares a technical report and presentation.**
  - **Seminar schedule prepared at the start of the semester and monitored by course coordinator.**
  - **Faculty evaluates using rubrics (content depth, presentation skills, technical clarity, use of references).**
- **Documentation**
  - **Seminar report submitted with references, case studies, and review of latest research papers.**
  - **Evidence maintained: *seminar schedule, presentation slides, evaluation sheets, faculty remarks*.**

2. Process of Mini/Micro Projects

- **Identification of Projects**
  - **Students identify mini/micro projects related to course modules, local problems, or lab work extensions.**
  - **Ideas reviewed by faculty mentors for feasibility and PO/PSO alignment.**
- **Execution & Monitoring**
  - **Duration: typically 6–8 weeks within a semester.**
  - **Faculty guide monitors progress through weekly reviews and logbooks.**
  - **Deliverables include a working model/prototype, technical documentation, and demo.**
- **Evaluation**
  - **Based on innovation, technical depth, design methodology, cost-effectiveness, and applicability.**
  - **Evaluation rubrics include problem identification, design, results, teamwork, and communication.**
  - **External examiner involvement (if possible) enhances credibility.**

3. POs/PSOs Addressed

- **Seminars primarily address:**
  - **PO10 (Communication): Oral and written technical communication.**
  - **PO12 (Lifelong Learning): Independent learning on emerging topics.**
  - **PO9 (Team Work): When seminars are team-based.**
  - **PSO1 / PSO2: Domain-specific knowledge sharing.**
- **Mini/Micro Projects primarily address:**
  - **PO2 (Problem Analysis): Identifying and analyzing small-scale problems.**
  - **PO3 (Design/Development): Developing solutions and prototypes.**
  - **PO4 (Investigations): Data collection, experimentation, validation.**
  - **PO5 (Modern Tool Usage): Using CAD/software/analysis tools.**
  - **PO7 (Environment/Sustainability): Projects with social or eco-friendly relevance.**
  - **PO9 (Team Work) & PO10 (Communication): Team-based execution and presentation.**
  - **PSOs: Discipline-specific application of knowledge (e.g., AI/ML coding, IoT prototypes, embedded circuits, etc.).**

4. Documentation & Evidence

- **Seminars:**
  - **Seminar schedule and guidelines.**
  - **Presentation slides, evaluation rubrics, faculty feedback sheets.**
  - **Seminar reports with literature review.**
- **Mini/Micro Projects:**
  - **Project proposal and approval sheet.**
  - **Progress reports/logbooks.**
  - **Prototype demonstration photos/videos.**
  - **Final evaluation reports with PO/PSO mapping.**

Marks Allocation (10)

- **Seminar Process & PO/PSO Mapping → 4 Marks**
- **Mini/Micro Project Process & PO/PSO Mapping → 4 Marks**
- **Documentation & Impact Evidence → 2 Marks**

2.5. Case Studies and Real-Life Examples (10)

1. Process of Integrating Case Studies & Real-Life Examples

- **Selection of Case Studies**
  - **Faculty identify case studies from industry practices, research articles, professional society resources (IEEE, ASME, IET), and government/NGO reports.**
  - **Focus on complex, open-ended problems rather than routine textbook problems.**
  - **Cases selected to be multidisciplinary, socially relevant, or industry-driven.**
- **Execution in Classrooms**
  - **Students are exposed to scenario-based learning where they analyze real situations.**
  - **Case discussions integrated into theory courses, labs, and project work.**
  - **Students present solutions/strategies and justify with reasoning, tools, and standards.**
- **Documentation & Evaluation**
  - **Faculty prepare teaching notes, problem background, and guiding questions.**
  - **Students assessed through presentations, written analysis, role plays, or group discussions.**
  - **Records maintained: *Case study files, student submissions, evaluation rubrics.***

2. Types and Complexity

- **Type of Case Studies Used**
  - **Industry-based: Product failures, process improvements, quality control.**
  - **Research-oriented: Biomedical data interpretation, AI/ML model outcomes, environmental impact analysis.**
  - **Societal issues: Waste management, renewable energy solutions, healthcare accessibility.**
  - **Entrepreneurial cases: Start-up models, cost–benefit analysis, financial feasibility.**
- **Complexity**
  - **Simple Level: Short cases for second-year students (basic design problems, ethical dilemmas).**
  - **Moderate Level: Multidisciplinary cases requiring analysis, data interpretation, and trade-offs.**
  - **High Level: Final-year cases linked with capstone projects, internships, or industry collaboration involving sustainability, safety, and ethics.**

3. POs Addressed

- PO1 (Engineering Knowledge): Applying theory to solve real-life problems.
- PO2 (Problem Analysis): Identifying and formulating industry/societal problems.
- PO3 (Design/Development): Developing alternative solutions with constraints.
- PO4 (Investigations): Data gathering, research methodology, and result analysis.
- PO6 (Engineer & Society): Assessing societal, legal, and cultural aspects.
- PO7 (Environment & Sustainability): Emphasis on eco-friendly and sustainable practices.
- PO8 (Ethics): Addressing ethical dilemmas in engineering solutions.
- PO9 (Individual & Team Work): Collaborative group discussions.
- PO10 (Communication): Presenting findings and solutions.
- PO12 (Lifelong Learning): Exposure to continuously evolving real-world problems.

4. PSOs Addressed (Examples by Discipline)

- CSE/IT: AI case studies, cybersecurity attacks, data analytics for decision-making.
- ECE/EEE: Smart grid case studies, IoT in healthcare, signal processing failures.
- Mechanical: Manufacturing defects, automotive design, energy systems analysis.
- Civil: Structural failures, water resource management, urban planning.

5. Documentation & Evidence

- Faculty-prepared case study repository (with teaching notes, source references).
- Student submissions (analysis reports, presentations, prototypes if applicable).
- Evaluation rubrics for assessing problem-solving, innovation, and application.
- Integration records showing case studies included in syllabus/course files.

Marks Allocation (10)

- Type & Complexity of Case Studies Integrated → 4 Marks
- POs/PSOs Mapping & Justification → 4 Marks
- Documentation & Evidence → 2 Marks

2.6. SWAYAM/NPTEL/MOOC/Self Learning (10)

1. Process of Promoting SWAYAM/NPTEL/MOOC/Self-Learning

- **Awareness & Orientation**
  - **Department organizes orientation sessions/workshops to introduce students to MOOC platforms (SWAYAM, NPTEL, Coursera, edX, Udemy, etc.).**
  - **Faculty mentors guide students to select courses aligned with curriculum and career goals.**
- **Registration & Allotment**
  - **Students register for courses individually or via NPTEL Local Chapter (if institution is a member).**
  - **Faculty coordinators keep records of enrollments.**
  - **Course selection linked with CO–PO–PSO mapping.**
- **Monitoring & Mentoring**
  - **Periodic progress monitoring (weekly/monthly) through dashboards or progress reports.**
  - **Faculty mentors track video lecture completion, assignments, quizzes.**
  - **Students submit learning logs summarizing key takeaways.**
- **Evaluation & Certification**
  - **Students complete final proctored exam (for NPTEL/SWAYAM) or online assessments (Coursera, edX, etc.).**
  - **Certificates collected and archived at department level.**
  - **Results integrated with internal assessments or as part of elective/credit transfer (as per university guidelines).**

2. Number of Students Registered & Certified

- **Records maintained for:**
  - **Number of students registered in each semester/year.**
  - **Number of students successfully certified (with % pass rate).**
- **Evidence:** ***Registration lists, certification copies, NPTEL portal reports, consolidated department records.***
- **Example Data:**
  - **CAY: 210 registered, 165 certified (78.6%).**
  - **CAYm1: 190 registered, 140 certified (73.6%).**
  - **CAYm2: 170 registered, 120 certified (70.5%).**

3. POs Addressed through MOOC/Self-Learning

- **PO1 (Engineering Knowledge): Expands theoretical foundation beyond syllabus.**
- **PO2 (Problem Analysis): Students exposed to case-based MOOC assignments.**
- **PO3 (Design/Development): Courses on product design, AI/ML, IoT, sustainability.**
- **PO4 (Investigations): MOOCs with research assignments and simulations.**
- **PO5 (Modern Tool Usage): Training in new tools, programming languages, cloud platforms.**
- **PO8 (Ethics): Professional courses stressing ethics in engineering/business.**
- **PO10 (Communication): Improved technical writing & presentation (via assignments/projects).**
- **PO12 (Lifelong Learning): Self-directed continuous learning — primary goal of MOOCs.**

4. PSOs Addressed (Examples)

- CSE/IT: AI, ML, Cybersecurity, Cloud Computing courses from NPTEL/Coursera.
- ECE/EEE: Embedded Systems, VLSI, Renewable Energy, Telecom Systems MOOCs.
- Mechanical: Additive Manufacturing, CFD, Automotive Design, Energy Systems.
- Civil: Smart Cities, Structural Analysis, Water Resource Management.

5. Documentation & Evidence

- Student enrollment lists with roll numbers.
- Certificates of successful course completion.
- MoU/registration with SWAYAM/NPTEL Local Chapter (if applicable).
- Consolidated report showing year-wise participation and success rate.
- Mapping sheet of course → POs/PSOs addressed → attainment evidence.

Marks Allocation (10)

- Awareness & Participation Process → 2 Marks
- Number of Students Registered & Certified (with evidence) → 4 Marks
- Mapping to POs/PSOs (justification) → 4 Marks

2.7. Solving Complex Engineering Problems Incorporating Sustainability Goals (20)

1. Concept & Scope

- **NBA emphasizes that engineering education must go beyond routine problem-solving and engage students in complex, open-ended problems that:**
  - **Involve multiple constraints (technical, economic, environmental, societal).**
  - **Require innovation, creativity, and multidisciplinary knowledge.**
  - **Directly contribute to the United Nations Sustainable Development Goals (SDGs).**
- **Complex problem-solving is integrated across multiple levels of the curriculum (core courses, projects, hackathons, and capstone).**

2. Core Courses Incorporating Sustainability

- **Project-Based Learning (PBL):**
  - **Core engineering courses redesigned to include mini-projects addressing real-world problems.**
  - **Example:** ***Thermal Engineering → Design of solar dryers (SDG 7: Affordable & Clean Energy).***
- **Problem-Based Learning (PrBL):**
  - **Students are presented with ill-defined, open-ended engineering challenges and asked to investigate solutions.**
  - **Example:** ***Civil Engineering → Urban flood management case study (SDG 11: Sustainable Cities).***
- **Integrated Design Projects (IDP):**
  - **Multidisciplinary projects within a course, combining theory + design + simulation.**
  - **Example:** ***IoT-based smart irrigation system (SDG 2: Zero Hunger, SDG 6: Clean Water).***

3. Mini/Micro Projects

- **Short-duration projects embedded in courses/labs.**
- **Students solve localized or scaled-down versions of complex problems.**
- **Example: *Design of a low-cost water filter for rural areas (SDG 6: Clean Water & Sanitation).***

4. Capstone Projects

- **Final-year projects with industry/real-world relevance.**
- **Encourage students to tackle multi-disciplinary, sustainable engineering challenges.**
- **Example: *Smart grids for renewable energy integration (SDG 7, SDG 13 Climate Action).***
- **Process: Proposal → DPRC Review → Monitoring (seminars, logs) → Prototype/Implementation → Final Evaluation → PO/PSO Attainment Mapping.**

5. Hackathons & Innovation Competitions

- **Encourage time-bound, team-based innovation on pressing social/engineering challenges.**
- **National events like Smart India Hackathon or institutional hackathons linked to SDGs.**
- **Example: *AI-based healthcare monitoring system (SDG 3: Good Health & Well-being).***

6. Other Activity-Based Learning

- **Workshops/Societal Projects: Students working with NGOs, municipalities, or industries to design sustainable solutions.**
- **Industry-Sponsored Challenges: Solving real-life engineering problems in collaboration with companies.**
- **Research Paper/Patent Outcomes: Projects leading to publications and patents in sustainable technologies.**

7. POs/PSOs Addressed

- PO2 (Problem Analysis): Identifying & formulating complex engineering problems.
- PO3 (Design/Development): Designing sustainable solutions under constraints.
- PO4 (Investigation of Complex Problems): Research, data analysis, experimentation.
- PO5 (Modern Tool Usage): Use of CAD, simulation, AI/ML, and other advanced tools.
- PO6, PO7 (Society & Sustainability): Assessing societal, health, and environmental impact.
- PO8 (Ethics): Ethical responsibility in sustainable design.
- PO9 & PO10: Teamwork and communication in collaborative projects.
- PO12: Lifelong learning through new methodologies and global challenges.
- PSOs: Discipline-specific outcomes (e.g., PSO1: Embedded systems, PSO2: Data analytics in CSE).

8. Documentation & Evidence

- Course files with PBL/PrBL lesson plans.
- Project proposals, monitoring reports, and prototypes.
- Hackathon participation records and outcomes.
- Attainment mapping sheets (CO → PO/PSO → SDGs).
- Student publications, patents, or awards linked to SDGs.

Marks Allocation (20)

- Integration in Core Courses (PBL/PrBL/IDP) → 6 Marks
- Mini/Micro/Capstone Projects → 6 Marks
- Hackathons & Other Activities → 4 Marks
- Documentation & SDG Evidence → 4 Marks

2.8. Steps Taken for Enhancing Industry Institute Partnerships (15)

1. Partial Delivery of Courses by Industry Experts

- **Guest Lectures & Module Delivery**
  - **Invite industry experts to deliver selected modules of core/elective subjects.**
  - **Example: *Cloud Computing course partly delivered by Microsoft Azure professionals.***
- **Benefits:**
  - **Bridges the gap between academic theory and industry practice.**
  - **Enhances exposure to latest technologies, tools, and industrial standards.**
  - **Directly contributes to PO12 (Lifelong Learning), PO5 (Modern Tool Usage), PSOs.**
- **Evidence: Timetables, circulars, lecture slides, attendance sheets, feedback forms.**

2. Industry-Supported Laboratories

- **Establishment of Labs/CoEs (Centers of Excellence)**
  - **Set up with industry funding, equipment, or software.**
  - **Examples:**
    - **Siemens Centre of Excellence (Mechanical).**
    - **Cisco Networking Academy Lab (ECE/IT).**
    - **TCS Data Analytics Lab (CSE).**
- **Benefits:**
  - **Hands-on experience with industry-grade tools.**
  - **Supports project-based learning and hackathons.**
  - **Promotes joint research/publications with industry.**
- **Evidence: MoUs, photos of lab setup, lab manuals, student projects executed in these labs.**

3. Industry-Offered Short-Term Programs & Training

- **Skill Enhancement Initiatives**
  - **Industry partners conduct short-term courses, certifications, and workshops.**
  - **Example: *AWS Academy courses on cloud deployment, Infosys Springboard certification in Agile Development.***
- **Duration: 2–12 weeks, conducted online/offline.**
- **Target Outcomes:**
  - **Upskilling in emerging domains (AI/ML, IoT, AR/VR, Cybersecurity).**
  - **Certifications aligned with industry expectations.**
- **Evidence: Brochures, student enrollment lists, certificates issued, feedback analysis.**

4. Internship & Industry Projects

- **Formal collaborations to ensure structured internships (4–8 weeks or semester-long).**
- **Capstone projects executed in partnership with industry (problem statements sourced from industry partners).**
- **Evidence: MoUs, project allotment lists, internship letters, evaluation reports.**

5. Faculty Development & Industry Immersion

- Faculty deputed for industrial training/immersion programs.
- Joint curriculum design workshops with industry experts to keep syllabi updated.
- Evidence: FDP reports, joint curriculum committee minutes.

6. Industry-Sponsored Research & Consultancy

- Joint research projects funded by industry.
- Consultancy services offered by faculty teams to solve real-life industrial problems.
- Evidence: Sponsored project sanction letters, consultancy MoUs, publications/patents.

7. Outcome & Impact

- Students gain employability skills, certifications, and project exposure.
- Faculty updated with latest technologies & industry practices.
- Strengthens Criterion 6 (Faculty Contribution), Criterion 5 (Student Performance), and Criterion 9 (Governance & Transparency).

Marks Allocation (15)

- Partial delivery of courses → 4 Marks
- Industry-supported labs → 4 Marks
- Industry short-term programs → 3 Marks
- Internships, projects, consultancy → 4 Marks

Comprehensive list of Documentation & Evidence

1. Partial Delivery of Courses (by Industry Experts)

- Invitations / Circulars issued for expert lectures.
- Course plan showing industry expert–delivered modules.
- Attendance records of students.
- Lecture materials / PPT slides provided by industry experts.
- Feedback forms (student + faculty).

2. Industry-Supported Laboratories / Centres of Excellence

- MoUs / Agreements with industry partners.
- Photos/videos of lab setup and inaugural events.
- Lab manuals prepared in collaboration with industry.
- Records of student projects/experiments carried out in the lab.
- Utilization records (number of students/faculty trained).

3. Industry-Offered Short-Term Programs / Training

- Brochures and schedules of training programs.
- Student registration and participation lists.
- Certificates of completion.
- Faculty mentoring reports.
- Feedback/impact analysis reports.

4. Internships & Industry Projects

- **Internship circulars and allotment letters.**
- **Industry-provided project problem statements.**
- **Internship logbooks and weekly reports.**
- **Certificates issued by host organizations.**
- **Final evaluation/viva reports.**

5. Faculty Development & Industry Immersion

- **Approval letters for faculty deputation to industry training.**
- **Reports submitted by faculty post-training.**
- **Evidence of faculty involvement in curriculum revision workshops.**

6. Sponsored Research & Consultancy

- **Project sanction letters and funding approvals.**
- **Consultancy MoUs and agreements.**
- **Deliverables (reports, software, prototypes).**
- **Joint publications or patents.**